

VATREEL · DESIGN DECISION RECORD

VatReel design for construction

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DOCUMENT CONTROL

Document ID	VTR-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/vatreel		
Source	docs/decisions/0002-design-for-construction.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Constructability review of the TRL 3 model; every change made under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted
- **Decided by:** Amish Chadha, by pre-approval: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

■ Context

Writing the prototype build plan (VTR-BLD-001) means every part must be makeable by the stated process and must fit and fasten to its neighbours (STANDARDS section 18; Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible"). The TRL 3 model was checked with build123d: no two components overlap, every joint face touches (a gap of 0.6 mm at most, a sliding fit), and nothing collides with the frame or the pit walls with the arms lifted, at the shallowest and at the deepest setting (`python cad/src/model.py --check` , 45 components). The changes below were needed to get there. None changes what VatReel does, its pitch or its safety case, except to make guarding more complete.

■ Decision

Table 1. Changes from the concept to the constructable design.

#	Component	The concept had	The constructable design has	Why
1	Arm frame	Two arms joined by a small tie tube 180 mm out from the pivot	Both arms welded to a 76.1 x 3 mm torque tube round the pivot line	The lift screw acts on the near arm only; the tie would have carried the far arm's moment in bending at about 280 MPa. The torque tube carries it in torsion at 29 MPa
2	Sprockets	12 teeth on the 30 and 40 mm shafts	15 and 30 teeth (chain 1), 17 and 51 teeth (chain 2); ratio kept at 6 to 1	A 12-tooth 08B sprocket cannot be bored to 30 or 40 mm
3	Chain 2 guard	A flat plate on the wall side	A closed case of 1.5 mm sheet round the chain, on two spacers	The plate left the chain open at its edges (R12)
4	Bearing tower	Open at the bottom; ratchet exposed	Bottom sheet closes the chain box; a cover over the ratchet and pawl with the release lever outside	Close the last openings to chain 1 and the pawl (R12)
5	Far bearing	A split block with a bolted cap	An open-topped saddle with a keeper that swings shut	The far bearing is 0.9 m into the pit; the reel must drop in without anyone reaching there
6	Near bearing	Solid housing	Split block with a bolted cap	The reel is fitted from above, not slid along its axle
7	Pivot tube support	A bracket from the stand round the chain 2 sprocket	A plug in the tube's near end riding on the jackshaft tip	No bracket can pass the sprocket and the swinging arm; the jackshaft carries it at 39 MPa
8	Arm frame location	Not set	An inner collar (inside the torque tube) and an outer collar on the pivot tube	Holds the frame along the tube so chain 2 stays in line
9	Pivot tube reach	One telescopic tube	Pivot tube 1.2 m plus a 1.55 m extension pinned inside it at 100 mm steps; far stand clamps on either through a split sleeve	Spans 1.0 to 2.52 m with no tube end standing out more than about 150 mm
10	Splash shield	1.12 m wide, from 260 mm above the rim	1.32 m wide, from 60 mm, with a hole for the jackshaft	Sight lines from the reel to the operator's eye missed the narrower shield (R9)
11	Base frame	1.55 m long	1.75 m long	Carries the wider shield
12	Tower braces	Welded to the tower	Bolted, and lowered to clear the ratchet cover	Keeps the tower at 23.2 kg to carry; the braces clashed with the cover
13	Spiders	Spokes butted to the axle	Spokes on a 40 mm hub sleeve, hexagon ties between them	A joint that can be jigged and welded square
14	Carrier bars	Bars welded to the spokes	Bars with end plates bolted to the spoke tips; eye tabs for snap hooks	Bars can be replaced; the reel is carried as one piece
15	Axle sprocket	Hub toward the arm	Hub on the wall side	The hub hit the guard case's inner plate
16	Loading hooks	Not sized	1.5 m hooks stored on the base frame's back rail	They must reach the far end of a carrier bar from the near rim

#	Component	The concept had	The constructable design has	Why
17	Reel installation	Not worked out	A 3.6 m assembly bar through the hollow axle; two people on opposite rims carry the reel onto the raised arms	No one reaches 0.9 m over the pit
18	Assembly order	Not worked out	The thirteen steps of VTR-BLD-001 section 4, set by what fits past what	The arm frame must be on the pivot tube before the tube goes onto the jackshaft

■ Consequences

- `cad/src/model.py` holds every change; STEP and STL are regenerated; the general arrangement VTR-DWG-001 moves to Rev P2; the concept media and the build plan pictures are regenerated from the model.
- The parts cost moves to USD 1,437 against the USD 1,500 value-engineering target (USD 63 under). The guard case, cover and wider shield added about USD 40; the BOM lists every part.
- VTR-CAL-001 v0.2 reruns every check on the constructable design; all stresses stay within yield / 1.5 at the proof load.
- R8 is not met as written (two two-person modules, setup about 29 minutes); this is recorded as a decided limit in VTR-DEC-001.
- `design_state: constructable` is set in `project.yaml`.

Safety: Changes 3, 4, 5 and 10 improve guarding and splash protection. The plan's safety stops (VTR-BLD-001 section 6) apply to every step.